Caption: A Comprehensive Measure of Well-Being

Project Description:

The Human Development Index (HDI) Predictor harnesses Machine Learning to estimate a country's level of human development, offering users a fast and interactive prediction experience. The platform uses four key indicators: Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita to predict the HDI score and classify the result into Very High, High, Medium, or Low Human Development.

Utilizing a trained Linear Regression model, the HDI Predictor processes user inputs to generate meaningful predictions and insights. Built with Python, Scikit-learn, Pandas, Flask, HTML, CSS, and data visualization libraries, the platform provides a seamless and user-friendly experience. By combining health, education, and living-standard indicators, the HDI Predictor helps students, researchers, and policymakers better understand the factors that influence a country's overall human development.

Scenarios

Scenario 1: A user enters high values for Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita. The machine learning model processes these values and predicts a high HDI score. The system classifies the result under the Very High Human Development category.

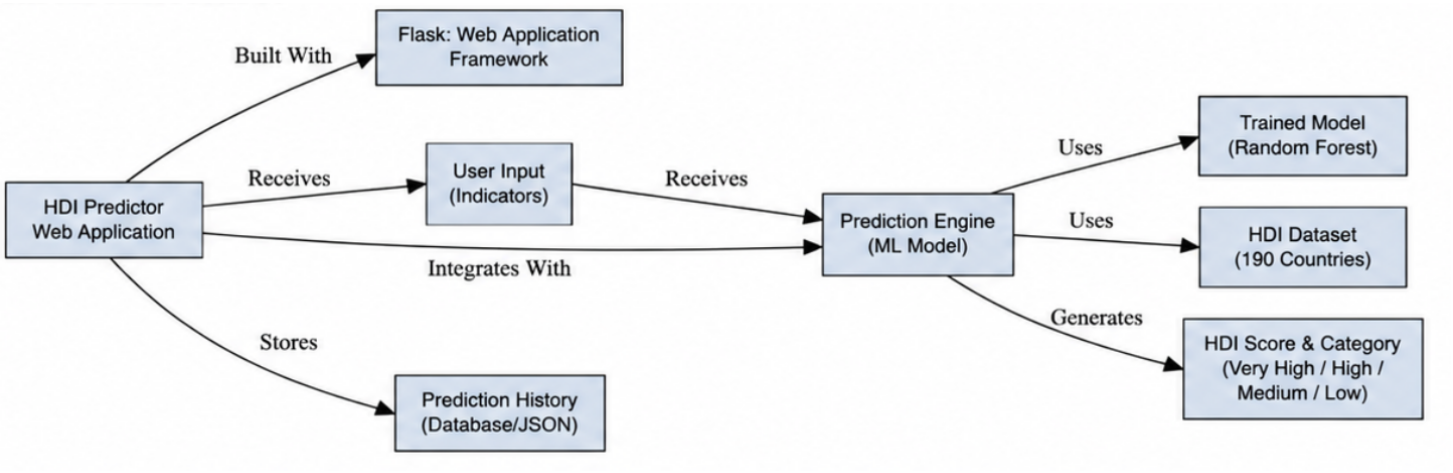
This scenario helps users understand how strong healthcare systems, better educational opportunities, and higher income levels contribute to improved human development.

Scenario 2: A policymaker or researcher enters moderate values for life expectancy, educational attainment, expected schooling, and income. The model predicts a Medium HDI score based on the provided indicators**.**

The result can help users understand the areas in which improvements are required. For example, improvements in healthcare, educational opportunities, or income generation may contribute to a higher level of human development.

Scenario 3 :A researcher enters relatively low values for Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita. The system predicts a Low HDI score and classifies the country under the Low Human Development category.
This prediction can help identify significant development challenges and highlight the need for improvements in healthcare, education, and economic opportunities.

Technical Architecture:



Description: The HDI Predictor follows a modular architecture consisting of a presentation layer, application layer, machine learning layer, and data storage layer.

The frontend is developed using HTML and CSS and provides a user-friendly interface through which users can enter the four required development indicators. The Flask backend receives the input values, validates and processes them, and sends the prepared values to the trained machine learning model.

The machine learning layer contains the trained Linear Regression model and the preprocessing components required to generate the HDI prediction. The predicted score is then converted into an appropriate human development category and displayed to the user.

The application also uses an SQLite database file for local data storage. The overall architecture ensures that the frontend, backend, machine learning model, and data storage components work together efficiently.

(Note: If you are using database in your project definitely you need to add ER Diagram along with description)

- ## Pre-requisites:
  - Flask Framework Knowledge: Flask Documentation
  - Machine Learning Familiarity: Scikit-learn Documentation
  - Data Analysis Skills: Pandas and NumPy Documentation
  - Data Visualization Knowledge: Matplotlib and Seaborn Documentation
  - HTML and CSS Skills: W3Schools HTML/CSS Tutorials
  - Python Programming Proficiency: Python Documentation
  - Development Environment Setup: Visual Studio Code and Python Installation Guide
  - SQLite Database Familiarity: SQLite Documentation

Project Workflow:

Milestone 1: Environment Setup and Package Installation

- **Activity 1.1:** Download and install all required Python packages, machine learning libraries, and Flask dependencies.
- **Activity 1.2:** Create and organize the complete project folder structure.

Milestone 2: Importing Required Libraries

- **Activity 2.1:** Import NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Pickle, Flask, and other required modules.
-

Milestone 3: Dataset Download and Understanding

- **Activity 3.1:** Obtain and load the HDI dataset.
- **Activity 3.2:** Understand the dataset structure, features, target variable, and data types.
- **Activity 3.3:** Perform exploratory data analysis and data visualization.

Milestone 4: Data Preprocessing and Label Encoding

- Activity 4.1: Select dependent and independent variables.
- Activity 4.2: Check and handle missing values.
- Activity 4.3: Encode categorical values where required.
- Activity 4.4: Prepare the processed dataset for model development.

Milestone 5: Dividing the Dataset into Train and Test Data

- **Activity 5.1:** Split the processed dataset into training and testing sets.

Milestone 6: Fitting the Model

- **Activity 6.1:** Train the Linear Regression model.
- **Activity 6.2:** Generate predictions.
- **Activity 6.3:** Evaluate the model using regression metrics and visualizations.

Milestone 7: Saving the Model

- **Activity 7.1:** Save the trained model.
- **Activity 7.2:** Store the required model files for future predictions.

Milestone 8: Building the Flask Web Application

- **Activity 8.1:** Develop the Flask backend.
- **Activity 8.2:** Create and integrate HTML templates.
- **Activity 8.3:** Run, test, and validate the complete application.

Milestone 1: Environment Setup and Package Installation

Milestone 1 focuses on preparing the development environment required to build the HDI Predictor. This includes checking the Python installation, creating the project structure, setting up a virtual environment, and installing all required packages.

Activity 1.1: Download and Install Required Packages

Before the application can communicate with the Groq LLM, you need a valid Groq API key. Follow the steps below to create one and connect it securely to the project.

• **Step 1: Verify Python Installation** :Open the terminal in Visual Studio Code and run:

python –version

The development environment used for this project contains:

Python 3.13.9

• **Step 2: Create and Activate the Virtual Environment**:

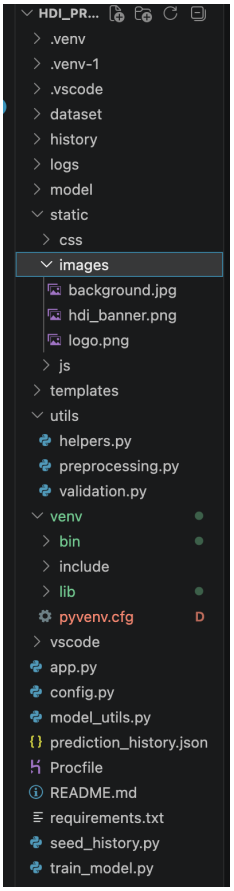
Create the virtual environment using:

python -m venv venv

Activate it using:

.\venv\Scripts\Activate.ps1

- **Step 3: Install Required Dependencies**
  Install all project dependencies using:
  pip install -r requirements.txt
  The required libraries support dataset processing, visualization, machine learning, model saving, database operations, and Flask web development.



Milestone 2: Importing Required Libraries

Activity 2.1: Import Required Libraries

In this milestone, all the necessary Python libraries and modules required for data analysis, visualization, machine learning, model development, model evaluation, database operations, and web application creation are imported.

The following libraries are used in the project:

- NumPy: Used for numerical calculations and array operations.
- Pandas: Used for loading, exploring, cleaning, and processing the HDI dataset.
- Matplotlib: Used for creating graphs and visualizing data patterns.
- Seaborn: Used for creating statistical visualizations and correlation heatmaps.
- Scikit-learn: Used for splitting the dataset, preprocessing data, training the Linear Regression model, generating predictions, and evaluating model performance.
- Pickle: Used for saving and loading the trained machine learning model and preprocessing components.
- Flask: Used for developing the web application backend, handling user requests, processing inputs, and displaying prediction results.
- SQLite: Used for local database storage and managing application data.

These libraries provide all the required functionalities for developing the complete Human Development Index Predictor, from dataset analysis and model training to prediction and web application integration.

app.py

app.py > ...

```python
"""
HDI Predictor Flask Application
"""

import webbrowser
from threading import Timer
from os import environ

from flask import Flask, render_template, request

from config import SECRET_KEY
from model_utils import predict_hdi
from seed_history import (
    save_prediction,
    load_history,
    initialize_history
)

app = Flask(__name__)
app.secret_key = SECRET_KEY

# Create history file if it doesn't exist
initialize_history()


def open_browser():
    webbrowser.open_new("http://127.0.0.1:5001")


@app.route("/")
def home():
    return render_template("index.html")


@app.route("/predict", methods=["POST"])
def predict():

    try:
        country = request.form["country"]

        life_expectancy = float(request.form["life_expectan
```

Milestone 3: Dataset Download and

Understanding Activity 3.1: Download and Load the Dataset

In this activity, the Human Development Index dataset is obtained and stored inside the dataset folder of the project. The dataset contains information related to the major indicators used to measure human development.

The dataset is loaded into the development environment using the Pandas library. Loading the dataset allows the data to be explored, analyzed, cleaned, and prepared for machine learning.

The main indicators used in the project are:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI per Capita

The target variable used for prediction is the HDI Score.

The dataset provides the foundation for training the machine learning model and understanding the relationship between health, education, income, and human development.

EXPLORER

HDI_PREDICTION FINAL P...

- .venv
  - bin
  - lib
  - .gitignore
  - .lock
  - CACHEDIR.TAG
  - pyvenv.cfg
- .venv-1
- .vscode
- dataset
  - Human Development I...
- history
- logs
- model
  - model_meta.json
  - model.pkl
- static
- templates
- utils
  - helpers.py
  - preprocessing.py
  - validation.py
- venv
  - bin
  - include
  - lib
  - pyvenv.cfg D
- vscode
- app.py
- config.py
- model_utils.py
- prediction_history.json
- Procfile
- README.md
- requirements.txt

OUTLINE

TIMELINE

Human Development Index - Full.csv

dataset > Human Development Index - Full.csv > data

```csv
Country,Life expectancy,Expected years of schooling,Mean years of schooling,GNI per capita,HDI
Afghanistan,65.5,15.8,13.0,19108,0.634
Albania,84.8,16.0,3.4,45924,0.677
Algeria,68.5,12.4,14.9,30444,0.679
Andorra,84.4,19.7,14.6,10470,0.729
Angola,64.6,17.8,8.8,43948,0.688
Antigua and Barbuda,70.7,19.5,13.0,11340,0.66
Argentina,69.4,10.8,7.8,18593,0.527
Armenia,80.3,11.4,13.4,45908,0.748
Australia,80.3,9.9,13.9,49145,0.752
Austria,75.1,15.0,3.3,26911,0.56
Azerbaijan,71.9,17.8,5.7,32026,0.628
Bahamas,73.6,8.8,8.9,70141,0.729
Bahrain,66.9,17.5,7.6,79571,0.811
Bangladesh,84.6,8.9,3.6,72755,0.708
Barbados,57.7,19.9,9.4,61217,0.758
Belarus,70.1,18.0,12.8,35976,0.733
Belgium,69.8,16.8,12.5,21528,0.662
Belize,72.2,14.3,12.7,73697,0.844
Benin,58.9,11.5,3.3,35402,0.5
Bhutan,80.6,14.3,12.7,75328,0.879
Bolivia,66.5,11.7,13.1,6680,0.551
Bosnia and Herzegovina,73.5,19.4,13.0,17616,0.693
Botswana,57.4,13.2,12.9,63210,0.744
Brazil,68.2,8.1,3.2,44373,0.531
Brunei,57.9,8.3,12.6,56800,0.668
Bulgaria,65.6,19.9,10.2,55845,0.776
Burkina Faso,70.8,8.0,14.8,24604,0.619
Burundi,70.4,15.8,8.7,30553,0.637
Cabo Verde,84.6,16.0,10.0,67890,0.847
Cambodia,62.0,16.2,4.1,12093,0.481
Cameroon,70.1,9.8,6.8,11498,0.479
Canada,63.7,14.6,5.2,66562,0.69
Central African Republic,57.1,16.7,9.9,49509,0.686
Chad,59.5,15.6,3.0,30437,0.52
Chile,72.6,15.8,12.4,38603,0.724
China,77.3,17.5,5.6,16591,0.585
Colombia,81.6,17.2,8.8,45106,0.746
Comoros,66.8,11.9,4.2,5672,0.432
Congo,67.3,15.8,13.6,5858,0.599
Costa Rica,67.9,9.2,4.7,63775,0.633
Cote d'Ivoire,75.9,9.8,11.1,58352,0.733
Croatia,55.3,10.9,7.1,62208,0.632
Cuba,63.2,13.8,4.3,41410,0.574
Cyprus,68.0,9.5,4.7,66670,0.648
```

Activity 3.2: Understand the Dataset

After loading the dataset, its structure is explored to understand the available features, target variable, number of records, data types, and statistical properties.

The following aspects of the dataset are examined:

- Number of rows and columns
- Column names
- Data types
- Missing values
- Statistical summary
- Feature distributions
- Target variable

The four main independent variables are Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita.

The dependent variable is the HDI Score.

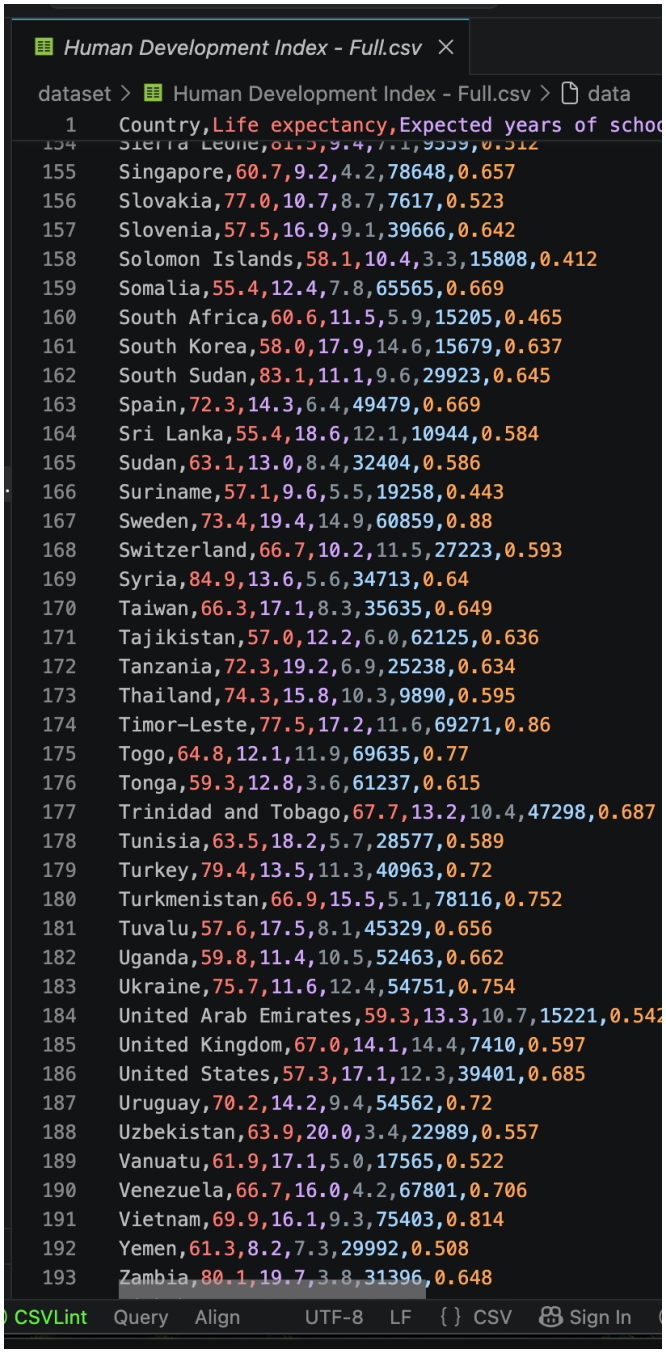
Understanding the dataset helps identify data quality issues and ensures that the correct variables are selected for model development.

Activity 3.3: Data Visualization

Exploratory Data Analysis is performed to understand patterns, trends, distributions, and relationships within the HDI dataset. The project includes different visualizations such as:

- Feature Correlation Heatmap
- HDI Score vs Life Expectancy
- HDI Score vs Mean Years of Schooling
- HDI Score vs Schooling Expected Years of
- HDI Score vs GNI per Capita
- Feature Distribution Plots

The correlation heatmap shows the relationship between the input features and the HDI score. Scatter plots help identify how changes in health, education, and income indicators are associated with changes in human development. These visualizations provide useful insights into the dataset before training the machine learning model.



Milestone 4: Data Preprocessing and Label Encoding

Activity 4.1: Select Dependent and Independent Variables

In this activity, the required input features and target variable are selected from the dataset. The independent variables used for prediction are:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI per Capita

The dependent variable is:

- HDI Score

The independent variables are provided to the machine learning model as input, while the HDI Score is used as the target output.

Selecting the correct variables ensures that the model learns the relationship between the important human development indicators and the final HDI score.

Activity 4.2: Check and Handle Missing Values

The dataset is checked for missing or null values before model training.

Missing values can reduce data quality and negatively affect model performance. Therefore, the

dataset is examined carefully to identify incomplete records.
Appropriate preprocessing techniques are applied wherever required to ensure that the dataset is clean and suitable for machine learning.

After

```python
@app.route("/about")
def about():
    return render_template("about.html")


if __name__ == "__main__":

    # Open browser automatically (only once)
    if environ.get("WERKZEUG_RUN_MAIN") == "true":
        Timer(1, open_browser).start()

    app.run(
        debug=True,
        host="0.0.0.0",
        port=5001
    )
```

preprocessing, the cleaned dataset is prepared for further model development.

Activity 4.3: Label Encoding
Machine learning algorithms require numerical data for model training. Therefore, categorical text values are converted into numerical representations wherever required.
The primary prediction target of the HDI Predictor is the numerical HDI Score. The predicted score is later interpreted into one of four human development categories:
- Very High Human Development
- High Human Development
- Medium Human Development
- Low Human Development

This classification makes the prediction easier for users to understand.

Activity 4.4: Prepare the Processed Dataset
After selecting the required variables, checking missing values, and processing the data, the dataset is prepared for machine learning.
The preprocessing stage ensures that the input data is clean, consistent, and suitable for model

training.
Feature scaling is applied where required to maintain consistent numerical ranges. The processed dataset is then used for train-test splitting and model development.

Milestone 5: Dividing the Dataset into Train and Test Data
Activity 5.1: Split the Dataset
In this milestone, the processed dataset is divided into training and testing sets.
The training dataset is used to train the machine learning model and help it learn the relationship between the four input indicators and the HDI score.
The testing dataset is used to evaluate the model using data that was not used during training.
The dataset is divided as follows:
- 80% of the data is used for training.
- 20% of the data is used for testing.

A random state is used to ensure consistent and reproducible results.
Dividing the dataset into training and testing sets provides a reliable method for evaluating the performance of the machine learning model on unseen data.

HDI_Predictor > tests > test_app.py > ...

```python
import os
import unittest
import sqlite3
import numpy as np
from app.predict import predict_hdi, classify_hdi_category
from app import utils

class TestHDIPredictor(unittest.TestCase):
    def setUp(self):
        # Con (variable) original_db: Literal['database.db']
        self.original_db = utils.DATABASE_PATH
        utils.DATABASE_PATH = "test_database.db"
        utils.init_db()

    def tearDown(self):
        # Restore the original database configuration and clean up
        utils.DATABASE_PATH = self.original_db
        if os.path.exists("test_database.db"):
            os.remove("test_database.db")

    def test_database_initialization(self):
        """Test database and tables creation."""
        conn = utils.get_db_connection()
        cursor = conn.cursor()
```

Milestone 6: Fitting the Model

Activity 6.1: Train the Linear Regression Model

In this activity, the Linear Regression algorithm is used to train the machine learning model. Linear Regression is selected because the target variable, HDI Score, is a continuous numerical value.

The model learns the relationship between the following input indicators:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI per Capita

The trained model uses these relationships to estimate the HDI score for new user inputs.

predict.py 3 | test_app.py 1 | Home - Human Development Index... | requirements.txt | .gitignore

HDI_Predictor > tests > test_app.py > ...

```python
class TestHDIPredictor(unittest.TestCase):
    def setUp(self):
        # Configure a temporary database path for testing
        self.original_db = utils.DATABASE_PATH
        utils.DATABASE_PATH = "test_database.db"
        utils.init_db()

    def tearDown(self):
        # Restore the original database configuration and clean up
        utils.DATABASE_PATH = self.original_db
        if os.path.exists("test_database.db"):
            os.remove("test_database.db")

    def test_database_initialization(self):
        """Test database and tables creation."""
        conn = utils.get_db_connection()
        cursor = conn.cursor()

        # Check users table
        cursor.execute("SELECT name FROM sqlite_master WHERE type='table' AND name='users'")
        self.assertIsNotNone(cursor.fetchone(), "Users table should exist.")

        # Check predictions table
        cursor.execute("SELECT name FROM sqlite_master WHERE type='table' AND name='predictions'")
        self.assertIsNotNone(cursor.fetchone(), "Predictions table should exist.")
```

Activity 6.2: Generate Predictions

After training the Linear Regression model, predictions are generated using the testing dataset.
The model receives the input features from the test data and generates predicted HDI scores.
The predicted values are compared with the actual HDI values to understand how accurately the model performs.
This process helps verify whether the trained model can generate reliable predictions for unseen data.

HDI_Predictor > model > create_notebook.py > ...

```python
import os
import json

# Setup notebooks directory
os.makedirs("notebooks", exist_ok=True)
notebook_path = os.path.join("notebooks", "HDI_Analysis.ipynb")

# Define cells structure
cells = [
    {
        "cell_type": "markdown",
        "metadata": {},
        "source": [
            "# Human Development Index (HDI) Exploratory Data Analysis & Modeling\n",
            "This notebook provides a comprehensive academic analysis of the Human Development Index (HDI) dataset,
        ]
    },
    {
        "cell_type": "markdown",
        "metadata": {},
        "source": [
            "## 1. Project Background and Indicators\n",
            "The Human Development Index (HDI) is a statistical measure compiled by the United Nations to evaluate t
            "- **Health**: Measured by Life Expectancy at Birth.\n",
            "- **Education**: Measured by Mean Years of Schooling and Expected Years of Schooling.\n",
            "- **Standard of Living**: Measured by Gross National Income (GNI) per Capita."
```

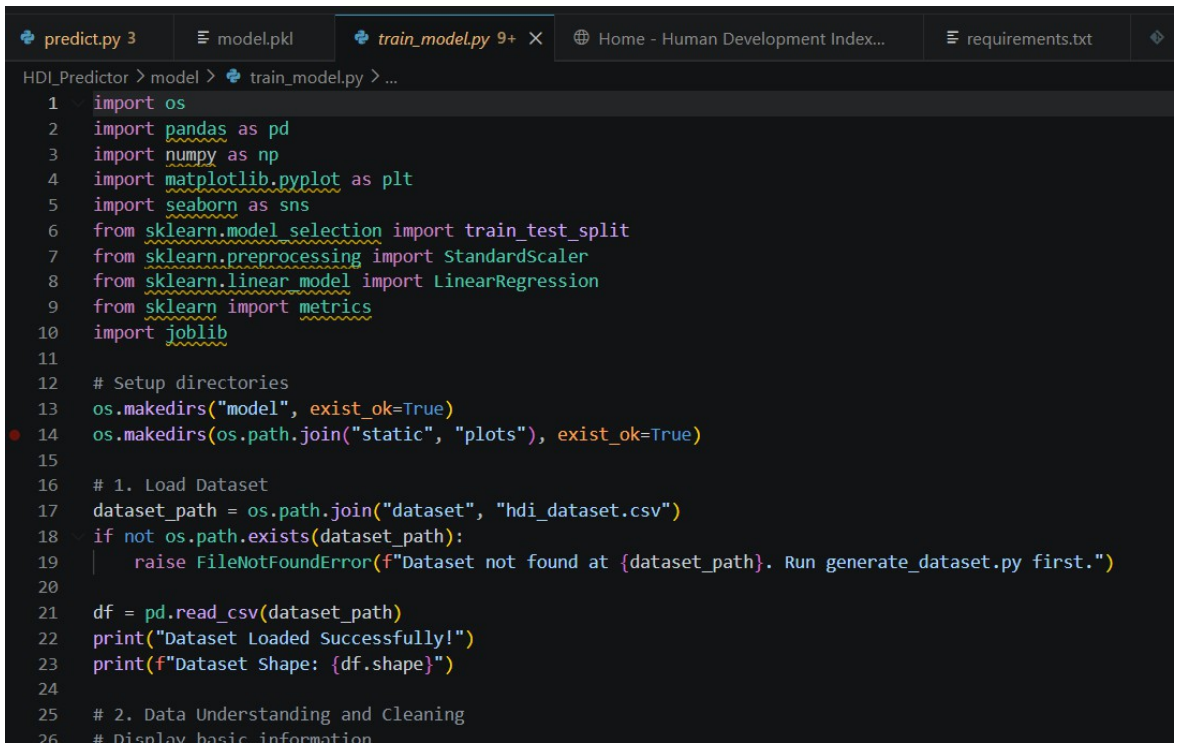
Activity 6.3: Evaluate Model Performance

The performance of the trained Linear Regression model is evaluated using regression evaluation metrics.
The following metrics are used:

- Mean Absolute Error (MAE): Measures the average absolute difference between actual and

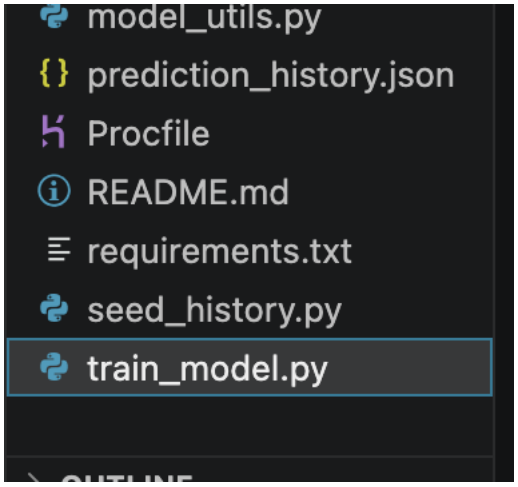
predicted values.
● Mean Squared Error (MSE): Measures the average squared difference between actual and predicted values.
● Root Mean Squared Error (RMSE): Represents the prediction error in the same unit as the target variable.
● R^2 Score: Measures how well the model explains the variation in the HDI score.

Milestone 7: Saving the Model
Activity 7.1: Save the Trained Model
After successful model training and evaluation, the trained Linear Regression model is saved using serialization.
Saving the model allows it to be reused by the Flask web application without training it again every time the application starts.
The trained model is stored inside the model folder of the project.
This improves application efficiency and separates the model training process from the prediction process.




```python
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn import metrics
import joblib

# Setup directories
os.makedirs("model", exist_ok=True)
os.makedirs(os.path.join("static", "plots"), exist_ok=True)

# 1. Load Dataset
dataset_path = os.path.join("dataset", "hdi_dataset.csv")
if not os.path.exists(dataset_path):
    raise FileNotFoundError(f"Dataset not found at {dataset_path}. Run generate_dataset.py first.")

df = pd.read_csv(dataset_path)
print("Dataset Loaded Successfully!")
print(f"Dataset Shape: {df.shape}")

# 2. Data Understanding and Cleaning
# Display basic information
```

Activity 7.2: Store Model Files for Future Prediction
The required machine learning components are stored so that they can be loaded whenever a user requests an HDI prediction.
The saved components support:
● Fast prediction
● Model reuse
● Reduced application startup time
● Separation of model training and web application execution
● Consistent preprocessing of user inputs
The saved model files are used by the Flask backend to process new inputs and generate HDI predictions.



Milestone 8: Building the Flask Web Application

Activity 8.1: Develop the Flask Backend

In this activity, the Flask backend is developed to connect the web interface with the trained machine learning model. The Flask backend performs the following operations:

- Starts the web application
- Defines application routes
- Displays HTML pages
- Receives user input
- Validates input values
- Processes the input features
- Loads the trained machine learning model
- Generates the HDI score prediction
- Classifies the HDI category
- Displays the prediction result

The backend acts as the main connection between the user interface, machine learning model, and database.

HDI_Predictor > venv > Lib > site-packages > flask > app.py > ...

```python
from __future__ import annotations

import collections.abc as cabc
import os
import sys
import typing as t
import weakref
from datetime import timedelta
from inspect import iscoroutinefunction
from itertools import chain
from types import TracebackType
from urllib.parse import quote as _url_quote

import click
from werkzeug.datastructures import Headers
from werkzeug.datastructures import ImmutableDict
from werkzeug.exceptions import BadRequestKeyError
from werkzeug.exceptions import HTTPException
from werkzeug.exceptions import InternalServerError
from werkzeug.routing import BuildError
from werkzeug.routing import MapAdapter
from werkzeug.routing import RequestRedirect
from werkzeug.routing import RoutingException
from werkzeug.routing import Rule
from werkzeug.serving import is_running_from_reloader
from werkzeug.wrappers import Response as BaseResponse
```

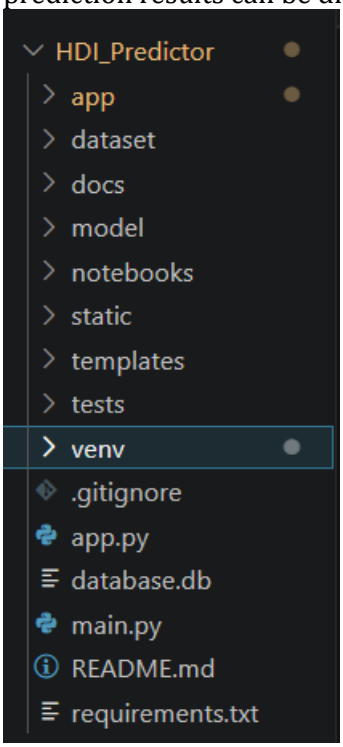
The user interface of the HDI Predictor is developed using HTML and CSS.

The templates folder contains the web pages required by the application.

The application includes:

- Home Page
- Predict HDI Page
- Prediction Result Page
- About Project Page

The static folder contains CSS files, images, visualization plots, and other resources required for the design of the application.
The HTML templates are integrated with Flask so that user requests can be processed and prediction results can be displayed dynamically.



Activity 8.3: Run, Test, and Validate the Application

After completing the backend and frontend integration, the complete Flask application is run and tested.
The application is started from the Visual Studio Code terminal.
The local web address displayed in the terminal is opened in a browser.
The application is tested by entering values for:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI per Capita

After entering the values, the user clicks the Predict button.
The Flask backend processes the inputs, sends them to the trained machine learning model, generates the HDI score, determines the human development category, and displays the result.
The complete application is validated to ensure accurate predictions and smooth functionality.

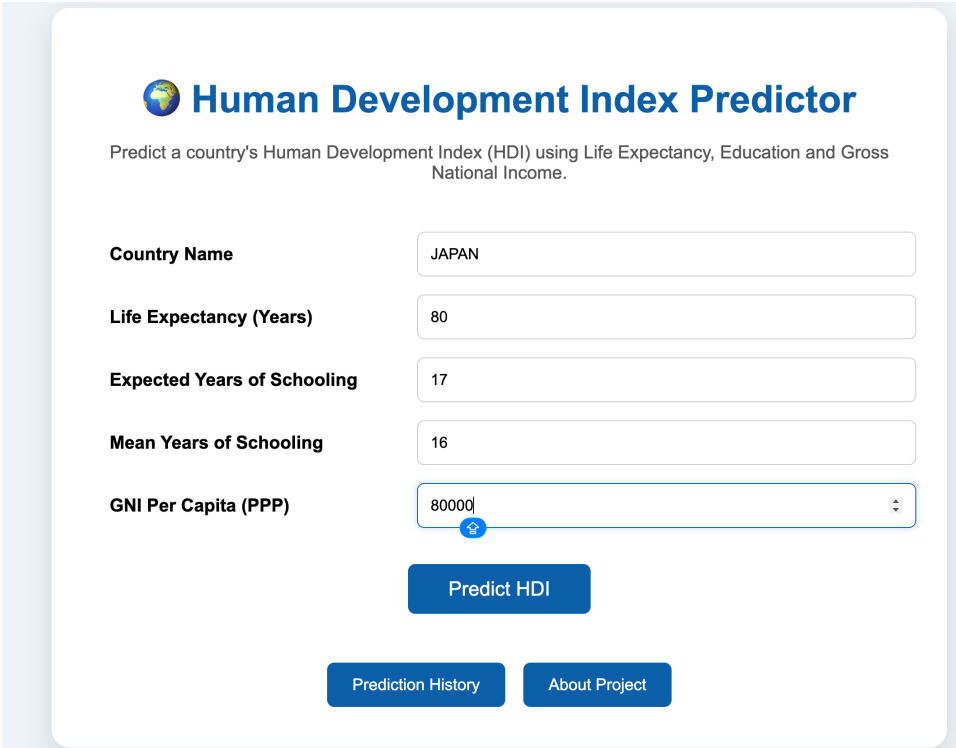
Exploring the Website's Web Pages

Home Page

The Home Page is the main landing page of the HDI Predictor. It introduces the concept of the Human Development Index and explains the purpose of the project.
The page highlights the four main indicators used by the prediction system:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI per Capita

The Home Page also provides an overview of the project and presents exploratory data analysis visualizations.



Four Dimension Parameters

The Four Dimension Parameters section explains the four major indicators used by the machine learning model.

- Life Expectancy: Represents the average number of years a newborn is expected to live under current mortality conditions. It reflects the overall health of the population.
- Mean Years of Schooling: Represents the average number of years of education received by people aged 25 years and older.
- Expected Years of Schooling: Represents the number of years of schooling that a child entering the education system can expect to receive.
- GNI per Capita: Represents Gross National Income per person and reflects the economic standard of living.

These four indicators collectively represent the major health, education, and living-standard dimensions of human development.

HDI Prediction Result

JAPAN

Indicator	Value
Life Expectancy	80.0 Years
Expected Years of Schooling	17.0 Years
Mean Years of Schooling	16.0 Years
GNI Per Capita (PPP)	$80000.0

Predicted HDI Score

0.889

Development Category

Very High Human Development

Predict Another Country | Prediction History | About Project

Exploratory Data Analysis Preview
The Exploratory Data Analysis section presents visual insights obtained from the HDI dataset.
The visualizations help users understand:
- Relationships between development indicators
- Correlations among features
- Distribution of values
- Factors associated with higher or lower HDI scores

The section includes visualizations such as the Feature Correlation Heatmap and HDI Score relationship plots.
These visualizations make the data easier to understand and provide useful insights into the factors that influence human development.

Predict HDI Page
The Predict HDI Page provides an interactive form where users enter the four required human development indicators.
The input fields include:
- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI per Capita

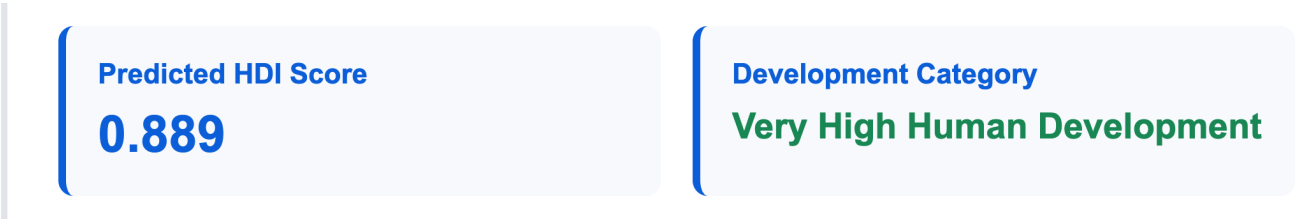
After entering the required values, the user clicks the Predict button.
The application sends the values to the Flask backend, where they are validated, processed, and passed to the trained machine learning model.

Indicator	Entered Value	Dimension
Life Expectancy	**78.5 Years**	Health
Mean Years of Schooling	**12.5 Years**	Education
Expected Years of Schooling	**15.6 Years**	Education
Gross National Income (GNI) per Capita	**$45,000.00**	Standard of Living

The Prediction Result Page displays the output generated by the trained machine learning model.
The page presents:
- Predicted HDI Score
- HDI Development Category

The predicted score is classified into one of the following categories:
- Very High Human Development
- High Human Development
- Medium Human Development
- Low Human Development

The result page provides a clear and user-friendly representation of the predicted human development level.



About Project Page

The About Project Page provides information about the purpose, importance, workflow, and technologies used in the HDI Predictor.
The page explains how machine learning is used to analyze health, education, and income indicators to estimate the Human Development Index.
It also helps users understand the objective of the project and the role of the four development indicators in measuring overall human well-being.

About Human Development Index Predictor

The Human Development Index (HDI) is a statistical composite index used to measure a country's overall development. It considers three major indicators:

- Life Expectancy
- Education
- Gross National Income (GNI) Per Capita

This application predicts the HDI score using a Machine Learning model trained on historical HDI data.

Development Categories

HDI Score	Category
0.800 - 1.000	Very High Human Development
0.700 - 0.799	High Human Development
0.550 - 0.699	Medium Human Development
Below 0.550	Low Human Development

Back to Home

CONCLUSION:-

The **Human Development Index (HDI) Predictor** project successfully demonstrates the application of **Machine Learning** and **Web Development** to estimate a country's Human Development Index based on four key development indicators: **Life Expectancy**, **Expected Years of Schooling**, **Mean Years of Schooling**, and **Gross National Income (GNI) per Capita**. The system provides users with an intuitive web interface for entering country information and receiving an accurate HDI prediction along with the corresponding human development category.

The application is developed using the **Flask** framework for the backend and **HTML, CSS, and JavaScript** for the frontend, while the prediction model is trained using **Scikit-learn's Random Forest Regressor**. The trained model is integrated into the web application using **Joblib**, enabling real-time predictions. Prediction history is also maintained to allow users to review previous results, making the application more practical and user-friendly.

The project demonstrates how artificial intelligence and predictive analytics can support researchers, students, policymakers, and development organizations in understanding the relationship between socio-economic indicators and human development. The system reduces manual analysis, provides quick insights, and offers a scalable platform for future enhancements.

Overall, the **HDI Predictor** is a reliable, efficient, and practical decision-support application that combines machine learning with modern web technologies to provide meaningful human development predictions. Future improvements may include incorporating additional socio-economic indicators, integrating live datasets from international organizations, providing graphical analytics dashboards, and deploying the application on cloud platforms for global accessibility.

THANKYOU